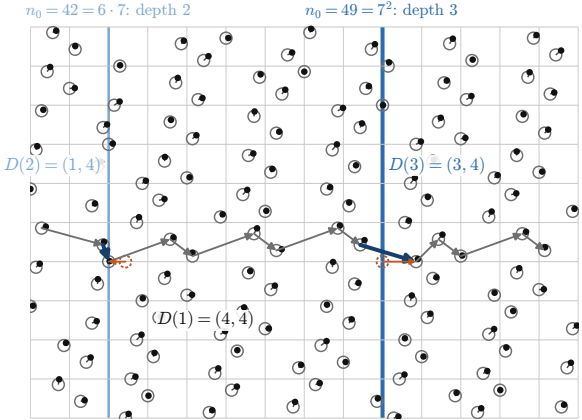
(a) $\mathrm{W}^-$, $p=7$, $P_6$: cells $n_0 \in [40, 54)$, $n_1 \in [100, 110)$ of $343 \times 343$;
every $x$-step changes $(r_0, r_1)$ by the column $D(l)$ of its boundary depth $l$




(b) carry responses
(the same at every level $K$)

| $l$ | $O_0^{(l)}$ | $X_{1\leftarrow 0}^{(l)}$ | $O_1^{(l)}$ | $X_{0\leftarrow 1}^{(l)}$ |
|---|---|---|---|---|
| 1 | 4 | 4 | 0 | 1 |
| 2 | 1 | 4 | 0 | 5 |
| 3 | 3 | 4 | 0 | 4 |
| 4 | 6 | 4 | 0 | 4 |
| 5 | 6 | 4 | 0 | 4 |
| 6 | 6 | 4 | 0 | 0 |
| 7 | 4 | 4 | 0 | 2 |
| 8 | 5 | 4 | 0 | 6 |

residues mod 7 ($4 = \frac{1}{2}$)

lattice image · actual point · jitter · depth $l=1$ · depth $l=2$ · depth $l=3$ · image if carry-free